

Rule Engine Feature & Library Documentation

This document provides a detailed mapping of features within the Rule Engine application to the libraries that power them.

1. Core Technologies

Library	Version	Purpose
React	^18.3.1	UI library for component-based architecture
Vite	^7.2.2	Fast build tool and dev server
Chakra UI	^2.10.0	Component library for styling and layout
Redux Toolkit	^2.11.0	State management for application data
Lucide React	^0.447.0	Modern icon set for dashboards
React Icons	^5.4.0	Supplemental icon packs
Framer Motion	^11.18.2	Animations and transitions
Axios	^1.7.7	HTTP client for API interactions

2. Feature-Specific Library Usage

Rule Builder

Allows users to create, test, and save custom logic rules for vehicle behavior. Libraries: Chakra UI, React Icons.

Simulation Dashboard

Central hub for running simulations and monitoring state. Libraries: Chakra UI, Lucide React, Framer Motion.

CAN Signal Builder & Code Generator

Defines CAN signals and generates MISRA-C compliant firmware code. Libraries: Chakra UI, React Icons.

Data Visualizer

Real-time visualization of vehicle data history using custom SVG charts.

Lifecycle & Trip Configuration

Configuration of device lifecycle states and trip parameters.

Hex Converter

Utility to convert raw CAN data into structured C-style arrays.

Payload Generation (Southbound)

Simulates communication with backend servers using structured JSON payloads.

3. Storage & Persistence

The Rule Engine uses a custom `useAutoPersist` hook leveraging `localStorage` and `Redux` to ensure simulation states persist across page refreshes. `Redux Persist` is optionally configured for advanced persistence needs.